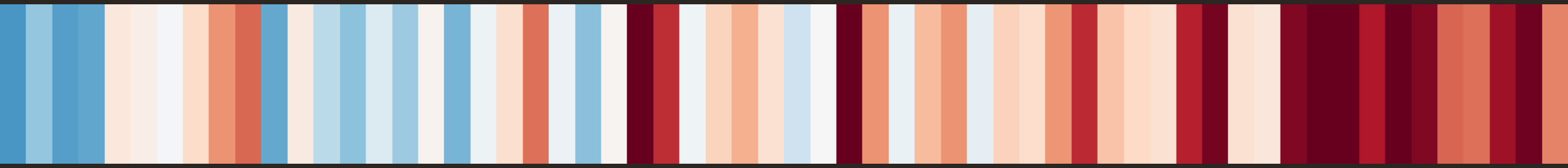
DATA ANALYSIS PROJECT

Is South Carolina Heating Up?

Three angles on 60 years of NOAA weather data (1966–2025)

[Your Name] · [Course name] · [Date]



1966 2025

WHY THIS TOPIC

Weather is global news, but people feel it locally

What has actually changed in South Carolina's weather over the last 60 years – **where, when, and how do people feel it?**

Answered with public NOAA data anyone can download, plus real statistics – not just eyeballing a chart.

1

Floods

Charleston floods at high tide, and the state had a "1000-year" flood in October 2015.

2

Heat

Summers are hot and humid – hot nights affect health and electricity bills.

3

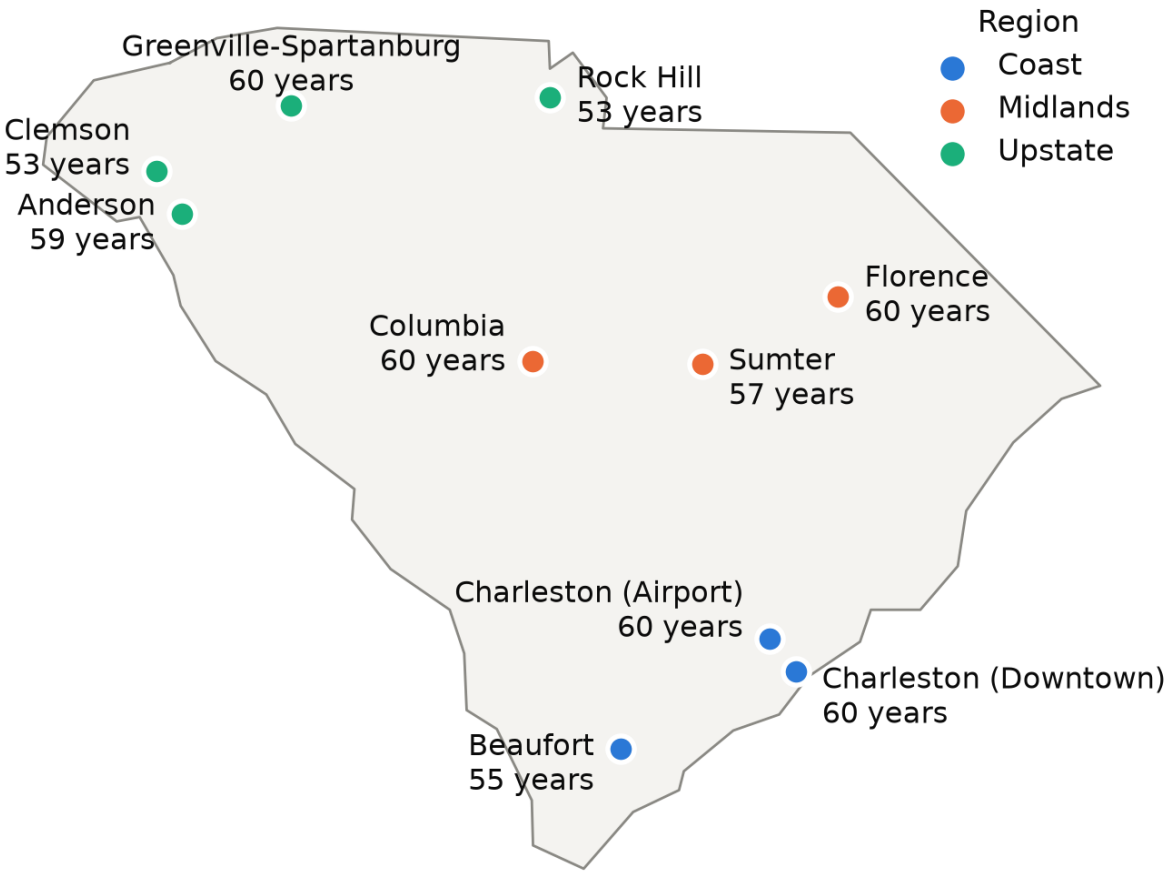
Farms

Agriculture depends on freeze dates and the length of the growing season.

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

TASK 1 · THE DATA

10 NOAA weather stations, 60 years, every single day



10
stations in 3 regions

60
years: 1966–2025

217,782
days of highs, lows and rain

Source: NOAA NCEI GHCN-Daily – the database behind NOAA's "Past Weather" page

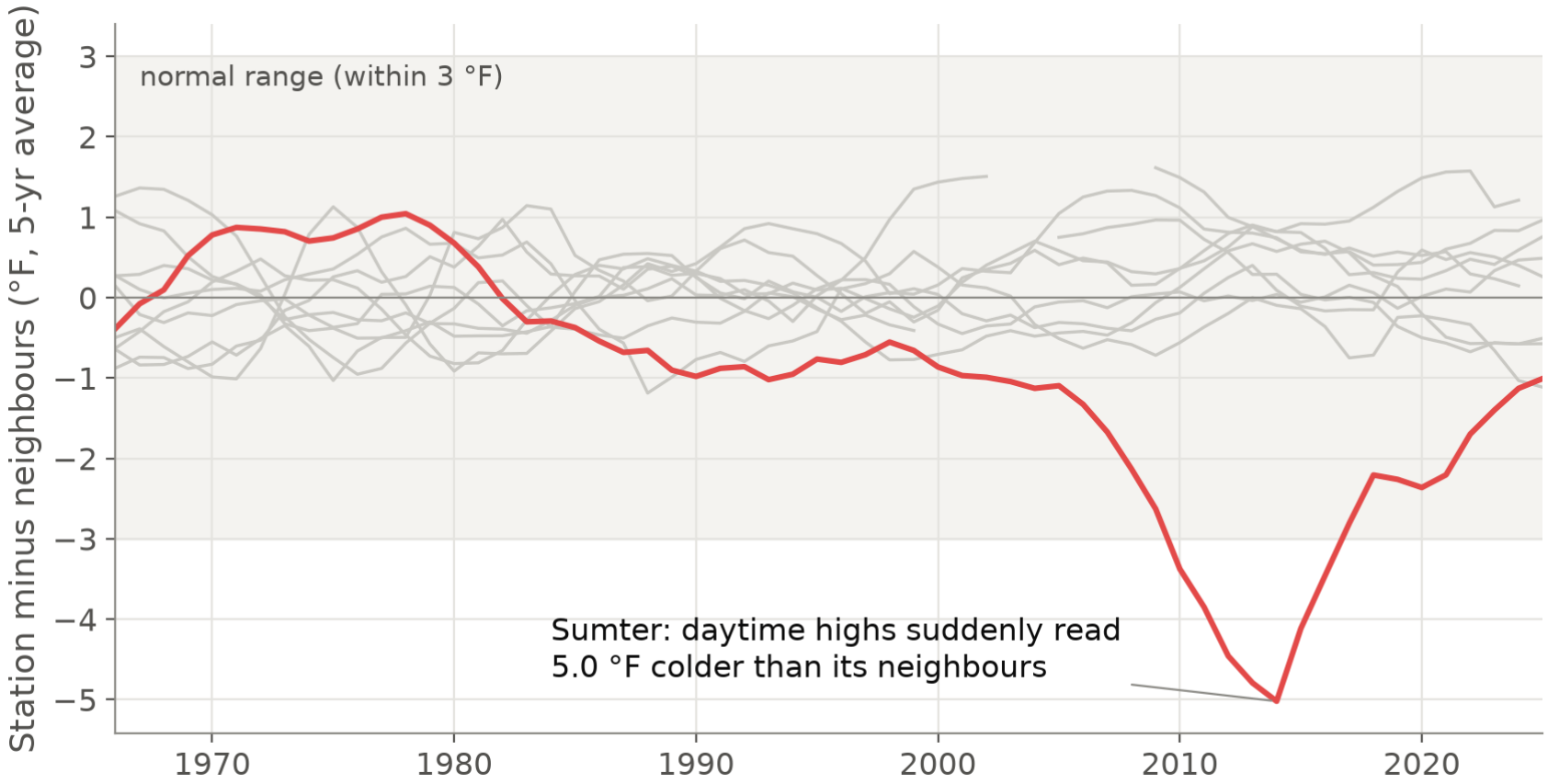
Download: one CSV per station from a public URL – no account or API key

Selection: 18 candidates → kept 10 with ≥ 330 days of data in (almost) every year

Variables: daily high (TMAX), daily low (TMIN), rain (PRCP)

TASK 2 · CLEANING

Cleaning the data – and catching a broken thermometer



−5 °F

Sumter's error vs. its neighbours (2010–2015) → left out of statewide averages

- Converted to °F and inches
- Removed values that failed NOAA's quality flags
- Kept only years with ≥ 330 days
- Neighbour check: grey = healthy stations

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

THE THREE ANGLES

One average can't tell the whole story – so three angles

1 WHERE

Which city has warmed the most?

Tells us if warming is local or statewide, and ranks cities for local planning. (The assignment's example question.)

2 WHEN

Day or night? Which season?

The yearly average hides how it warms. Night vs. day points to causes like city heat; seasons matter for crops and energy.

3 WHAT PEOPLE FEEL

Are extreme days more common?

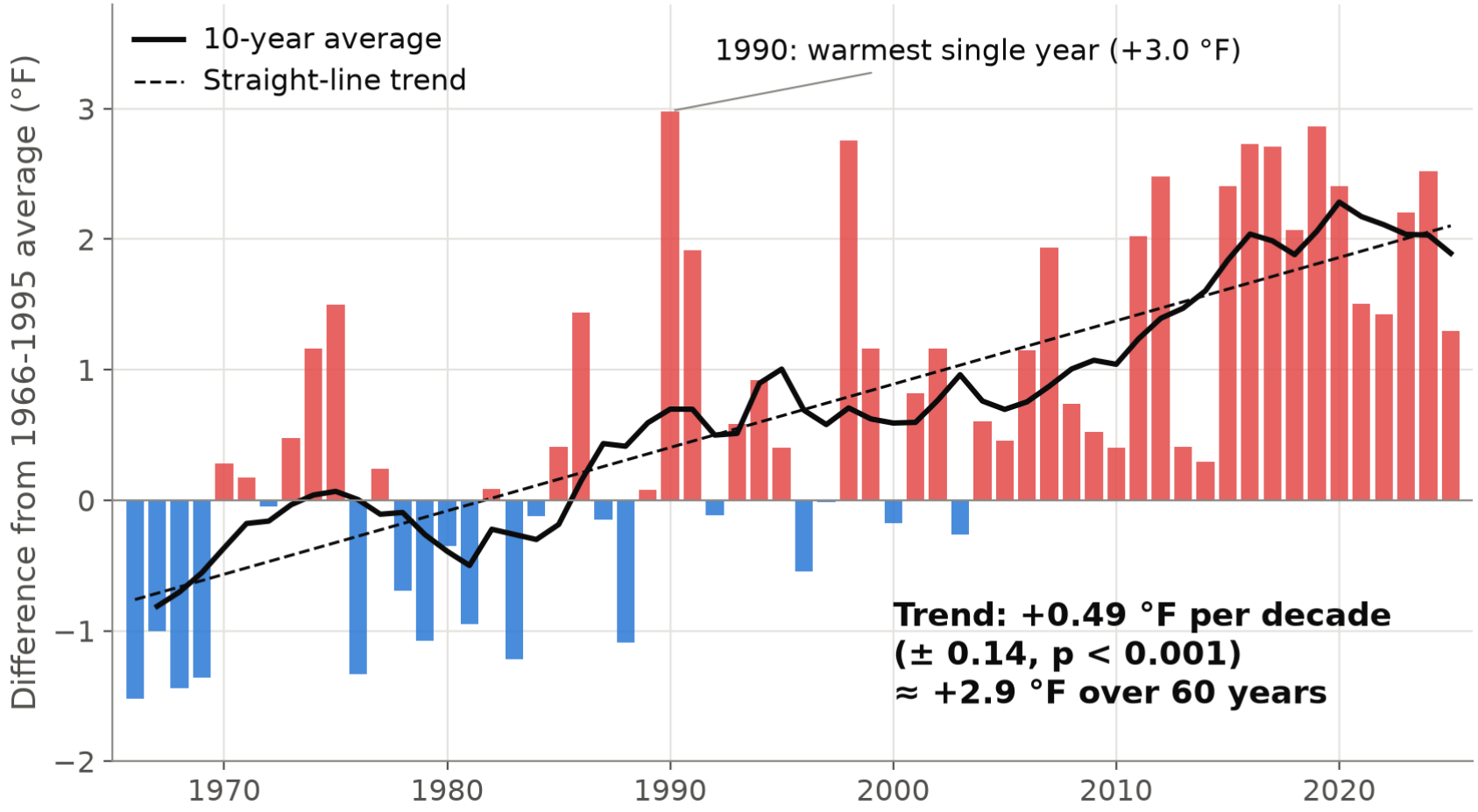
People feel thresholds – a 95 °F afternoon, a night that never cools, a frost, a flood. Also tests rain, a 2nd variable.

the average → **its timing** → **the extremes** · same cleaned data, same statistics: straight-line trend + p-value, and 1966–1995 vs. 1996–2025

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

ANGLE 1 · WHERE

South Carolina has warmed about 3 °F since the 1960s



+2.9 °F

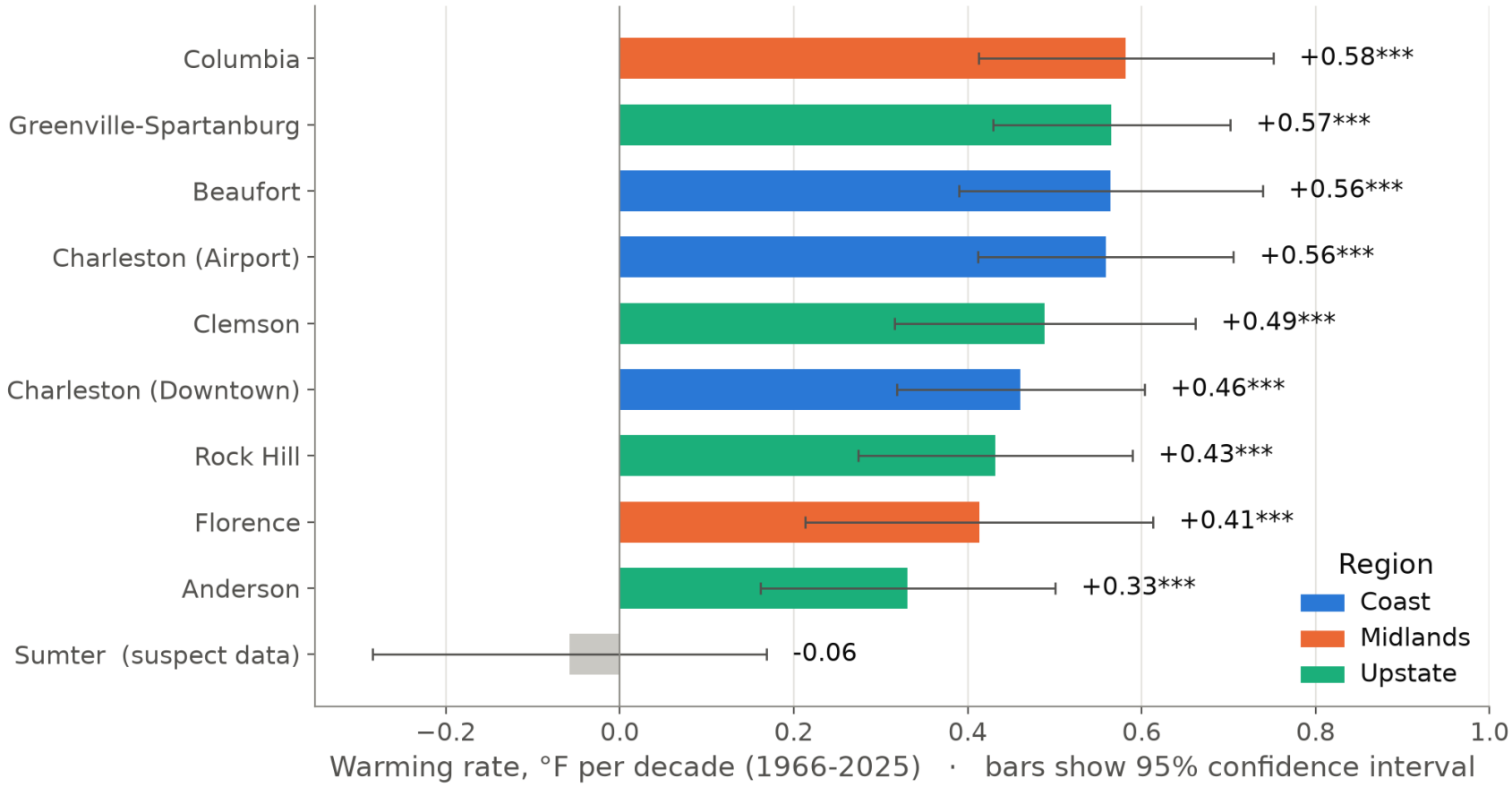
over 60 years · +0.49 °F per decade ($p < 0.001$)

- Before 1985 most years are blue; since 2010 almost all are red
- 8 of the 10 warmest years came after 2010
- Average of the 9 trusted stations

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

ANGLE 1 · WHERE

Columbia warms fastest – but the top four are a tie



+0.58

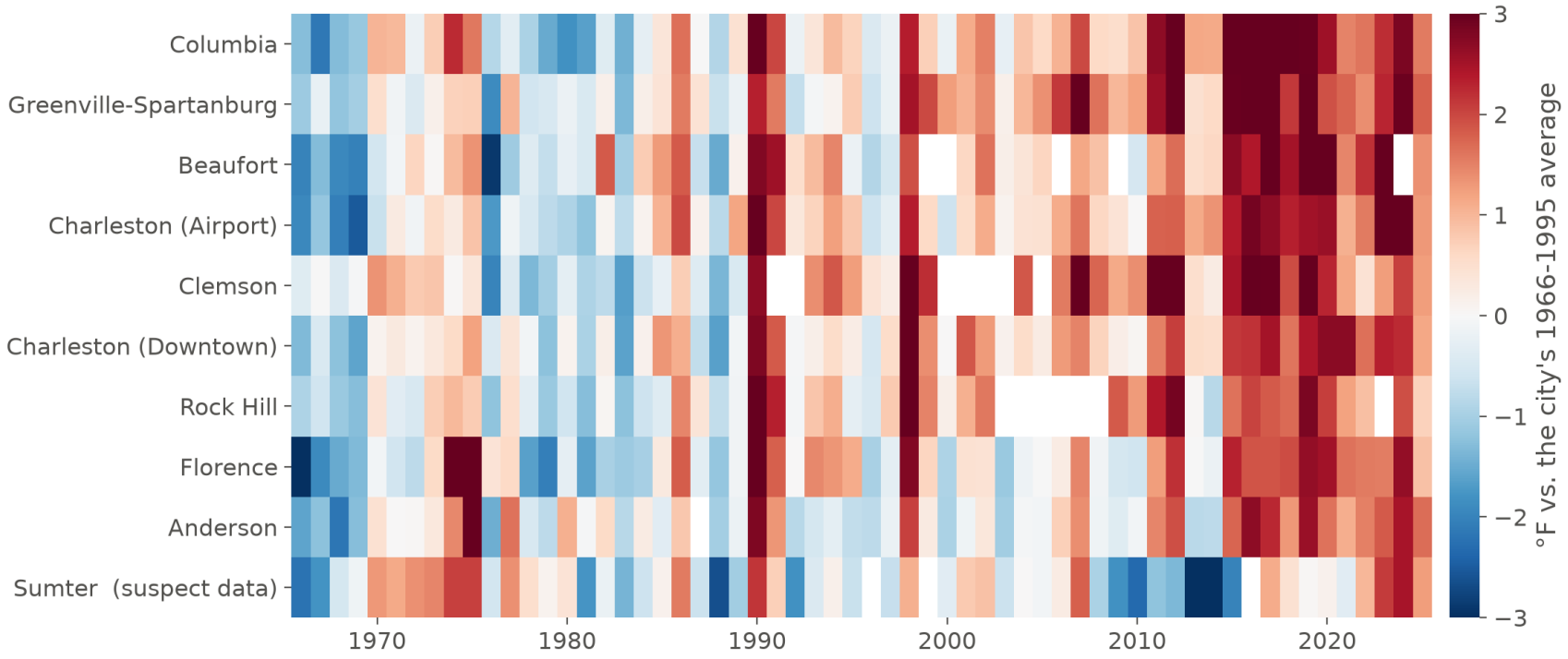
°F per decade in Columbia – about +3.5 °F since 1966

- Top 4 within 0.03 °F/decade – error bars overlap
- All 9 trusted stations warmed ($p < 0.001$)
- Coast 0.53 · Midlands 0.50 · Upstate 0.45

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

ANGLE 1 · WHERE

Warming shows up everywhere, at the same time



9 / 9

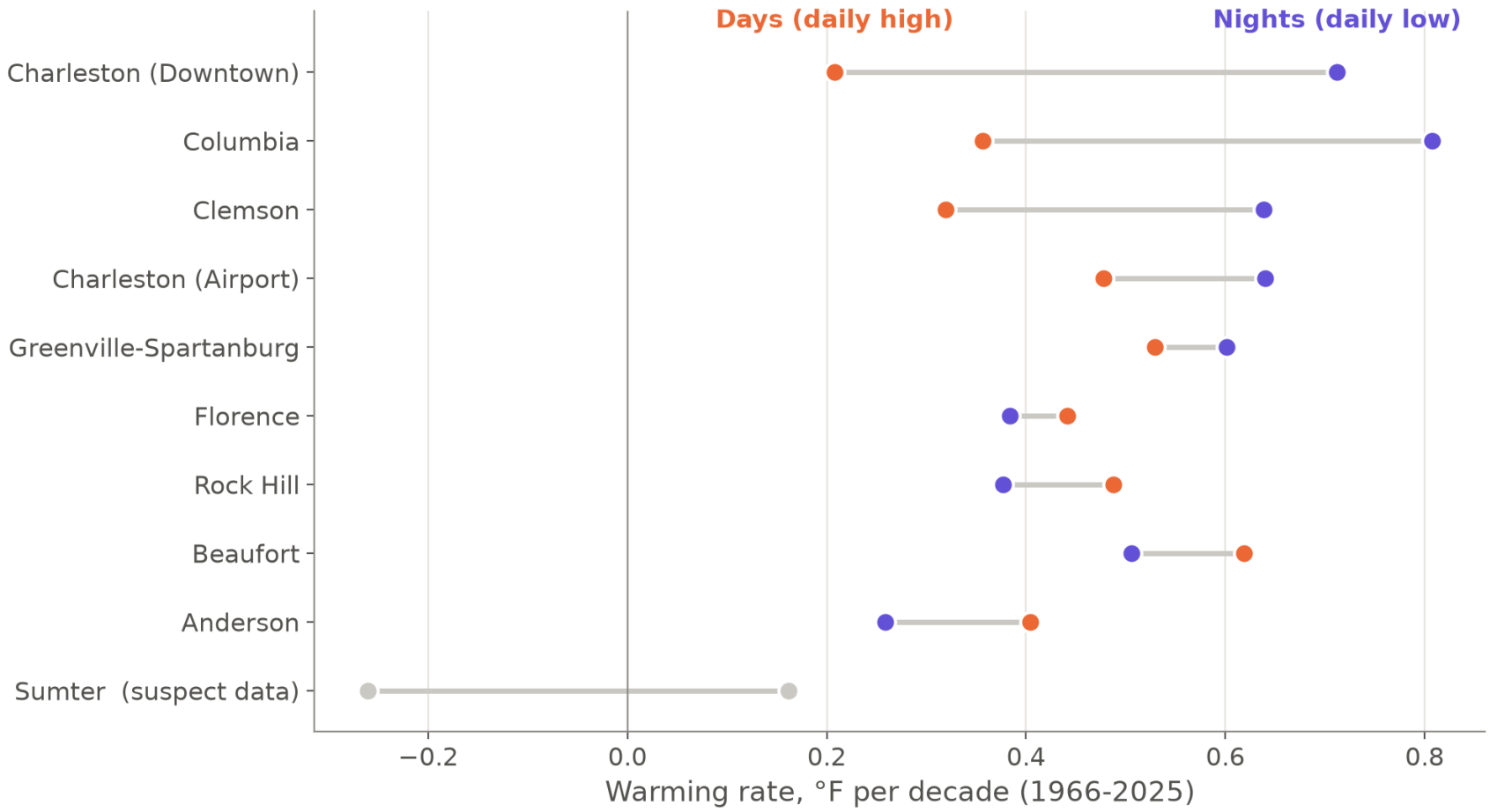
trusted stations have a significant warming trend

- Each row = a city, each column = a year
- Red = warmer than that city's own 1966–95 normal
- Sumter's broken years show as dark blue (2013–14)

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

ANGLE 2 · WHEN

Nights warm faster than days – most of all in cities



3×

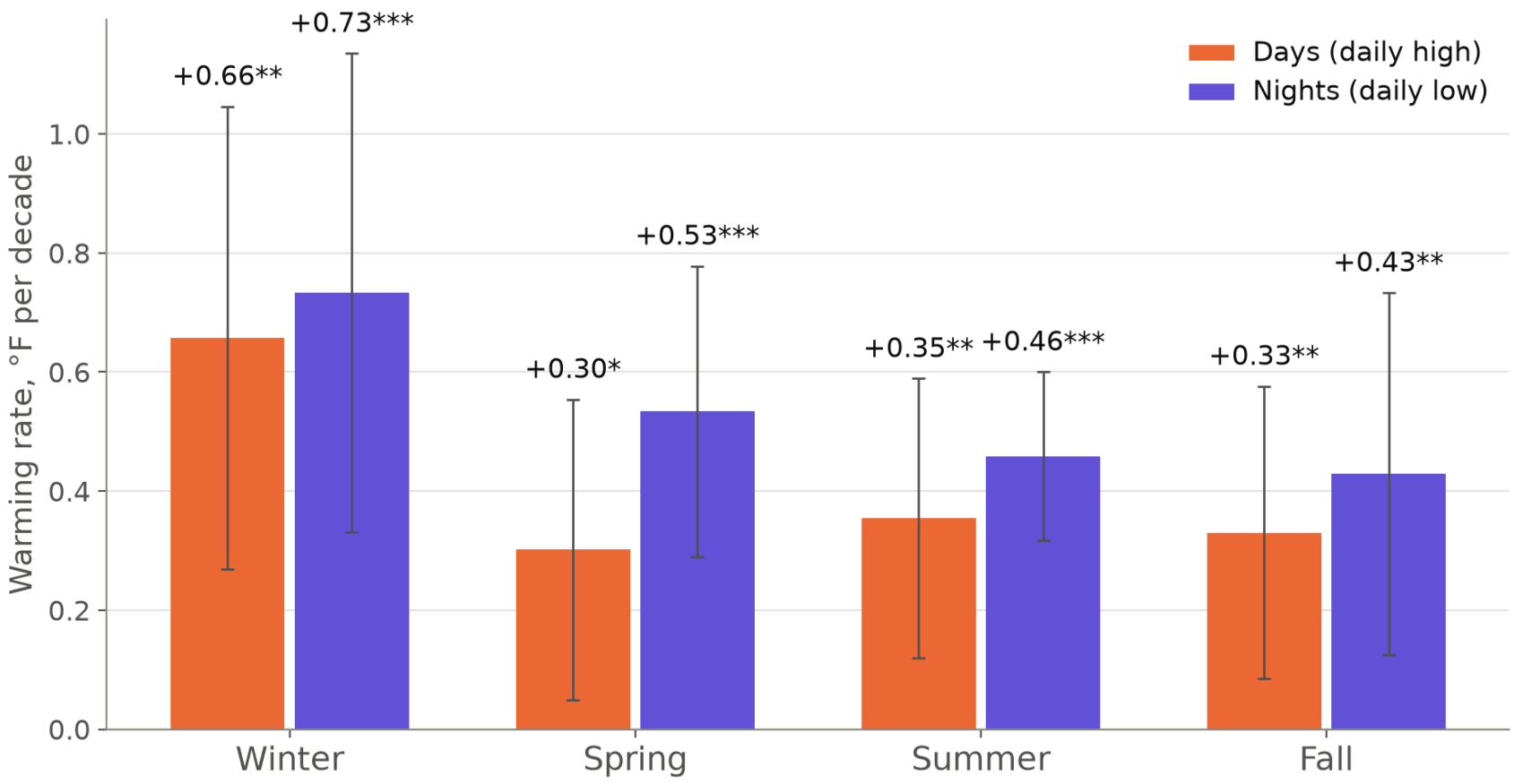
faster warming at night than by day in downtown Charleston

- Average: nights +0.55 vs. days +0.43 °F/decade
- Biggest gaps: downtown Charleston and Columbia
- Urban heat island: concrete stores heat, releases it at night

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

ANGLE 2 · WHEN

Winter is warming fastest



1.7×

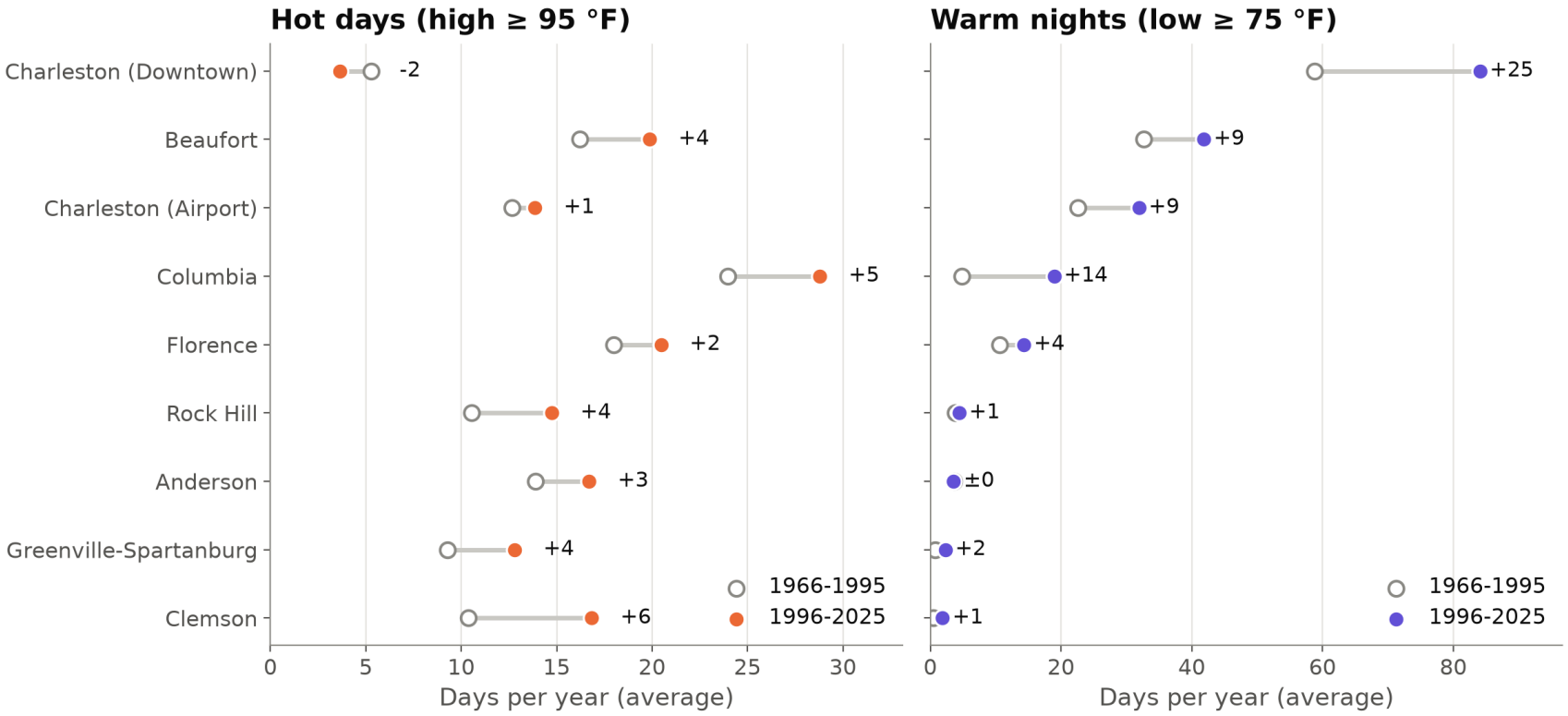
winter warms 1.7 times faster than the other seasons

- Winter: +0.66 (days), +0.73 (nights) °F/decade
- Other seasons: nights warm faster than days
- Winter varies a lot year to year → wide error bars

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

ANGLE 3 · WHAT PEOPLE FEEL

Many more nights that never cool below 75 °F



+47%

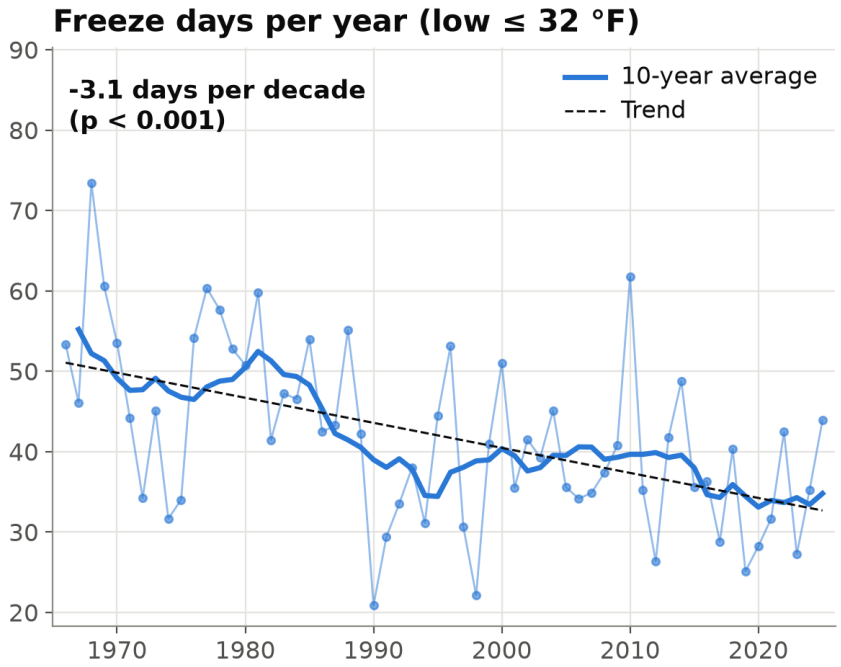
more warm nights statewide (15.6 → 22.9 per year)

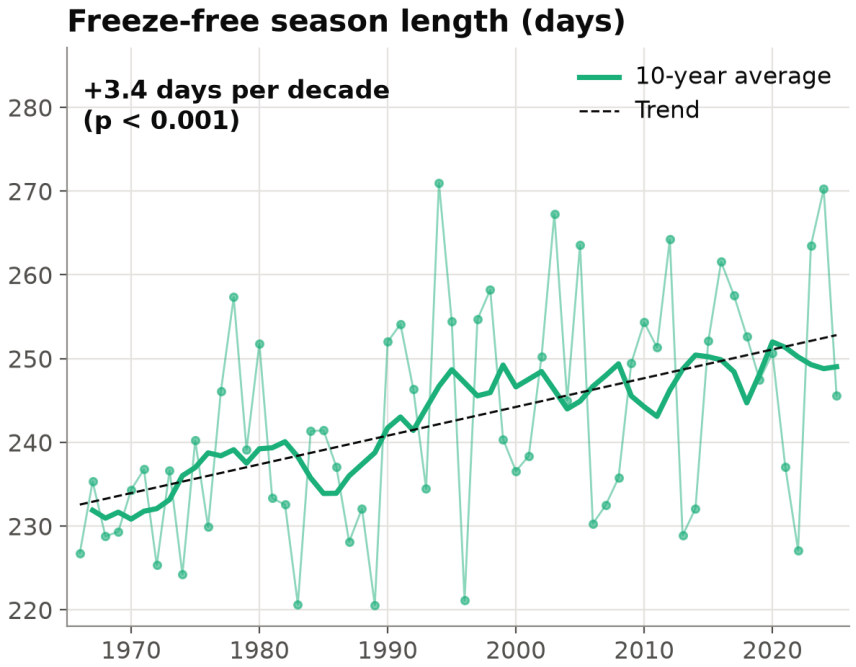
- Downtown Charleston: 59 → 84 warm nights a year
- Columbia: 5 → 19 (nearly 4×)
- Hot days ≥ 95 °F: +21% (just significant)

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

ANGLE 3 · WHAT PEOPLE FEEL

Softer winters: fewer freezes, a longer growing season





+3 weeks

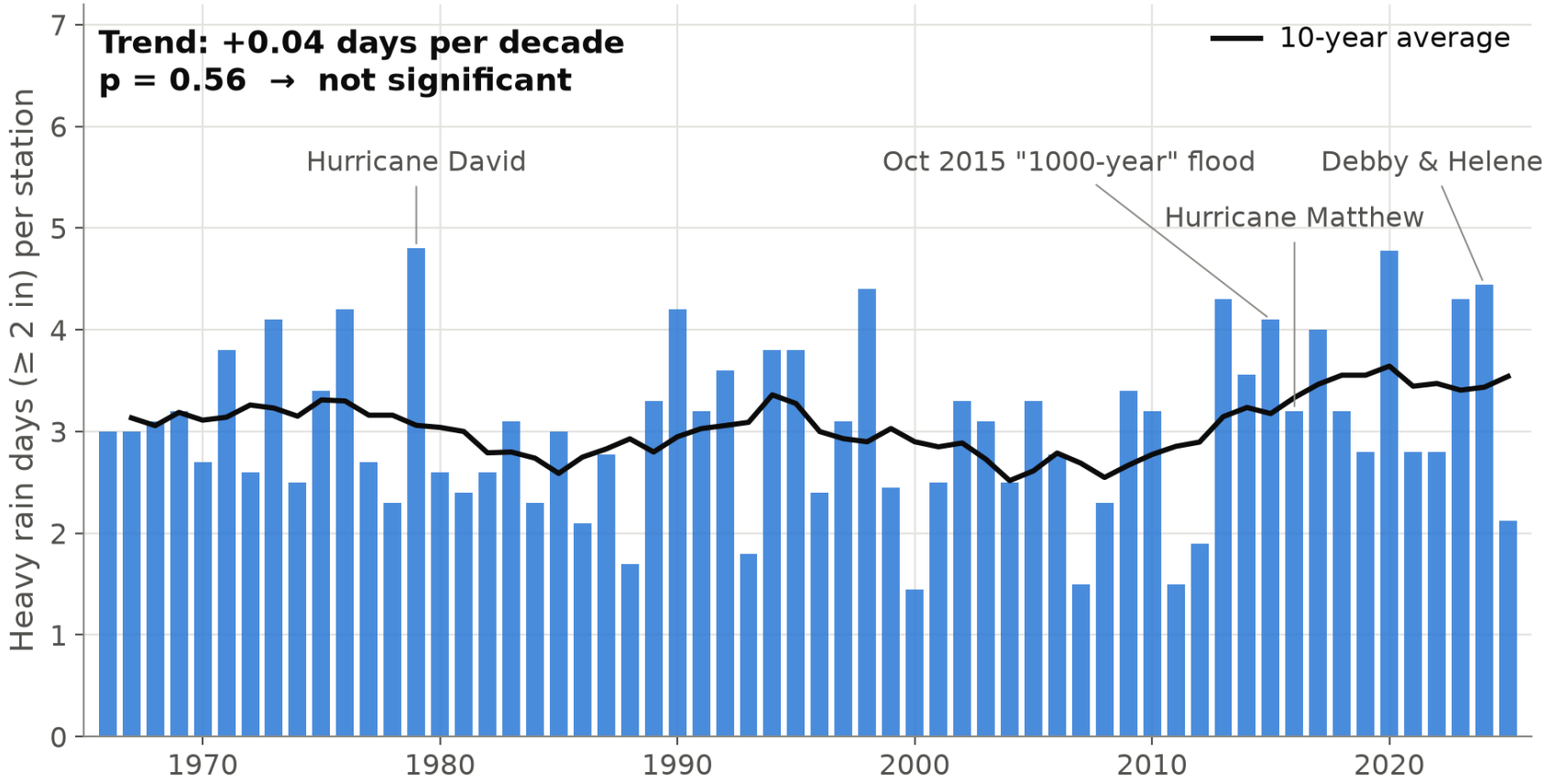
longer freeze-free (growing) season than in the 1960s

- Freeze days: 46 → 38 per year (−18%)
- −3.1 freeze days per decade (p < 0.001)
- Good for some crops – but also pests and pollen

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

ANGLE 3 · WHAT PEOPLE FEEL

The surprise: heavy rain shows no clear trend (yet)



p = 0.56

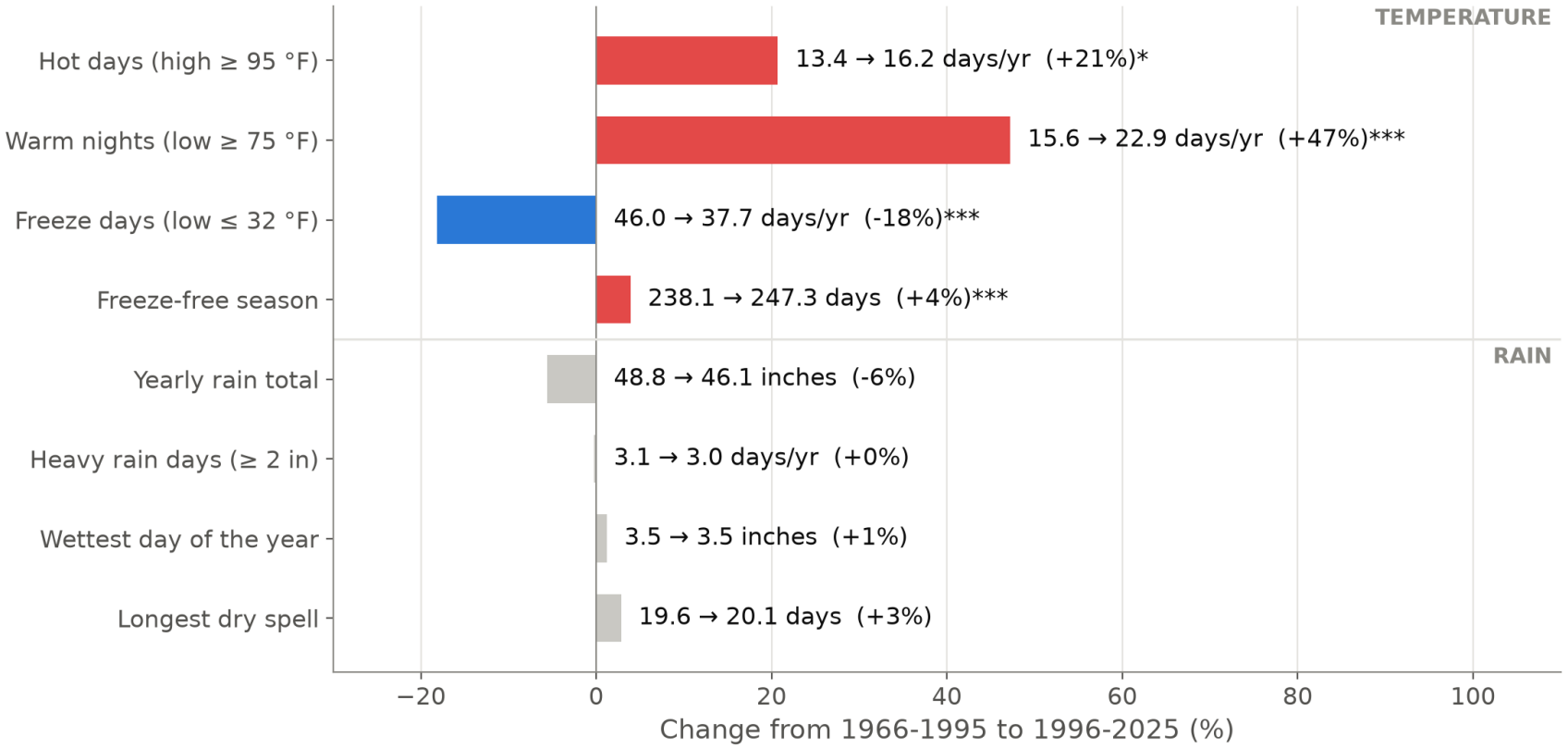
trend in heavy-rain days (≥ 2 in) is not significant

- Spikes are single storms: David, 2015 flood, Matthew, Debby & Helene
- Big storm years in every decade
- Rising since 2010 – worth watching

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

ANGLE 3 · WHAT PEOPLE FEEL

Scorecard: temperature extremes moved, rain did not



4 of 4

temperature extremes changed significantly – 0 of 4 rain measures did

- Coloured = significant 60-year trend
- Grey = could be natural ups and downs
- Rain is noisier – needs more data

Data: NOAA NCEI GHCN-Daily · 10 SC stations · 1966–2025

What I found

WHERE

+2.9 °F since 1966 at every trusted station. Columbia fastest (+0.58 °F/decade), tied with 3 others.

WHEN

Warming is strongest in winter and at night – nights warm fastest in cities.

WHAT PEOPLE FEEL

+47% warm nights, −18% freezes, 3-week longer growing season. Heavy rain: no clear trend yet.

Limitations raw station data with a simple neighbour check · cities grew around some stations · a straight line is a simple model · only 10 stations

Lesson check your data – one broken thermometer in Sumter would have changed the answer

Thank you – questions?

Everything is reproducible: python run_all.py rebuilds every chart in about 20 seconds

Code & data: [GitHub repository link]

Report: report/Project_Report.pdf

Data: NOAA NCEI GHCN-Daily (ncei.noaa.gov)

